Plain weave · plain_weave_compacted_high_vf.yaml · b3_tex 0.1.0

RVE settings

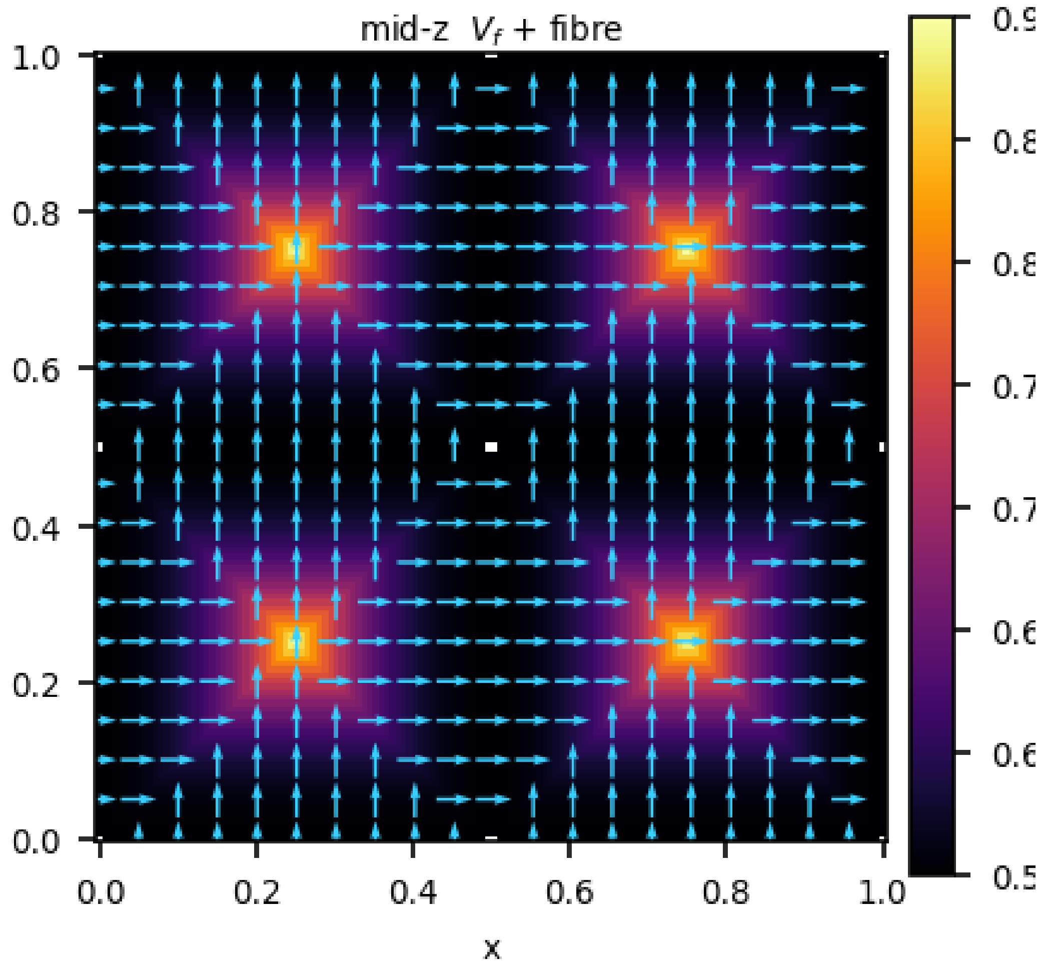
RVE size [m]	1 x 1 x 0.092
matrix material	matrix
yarn material	yarn
field type	woven
pattern	plain
n_warp	2
n_weft	2
warp width	0.49
warp height	0.076
section power	4.0
compaction	0.4
nominal Vf	0.55
max Vf	0.9
crossover nesting	on

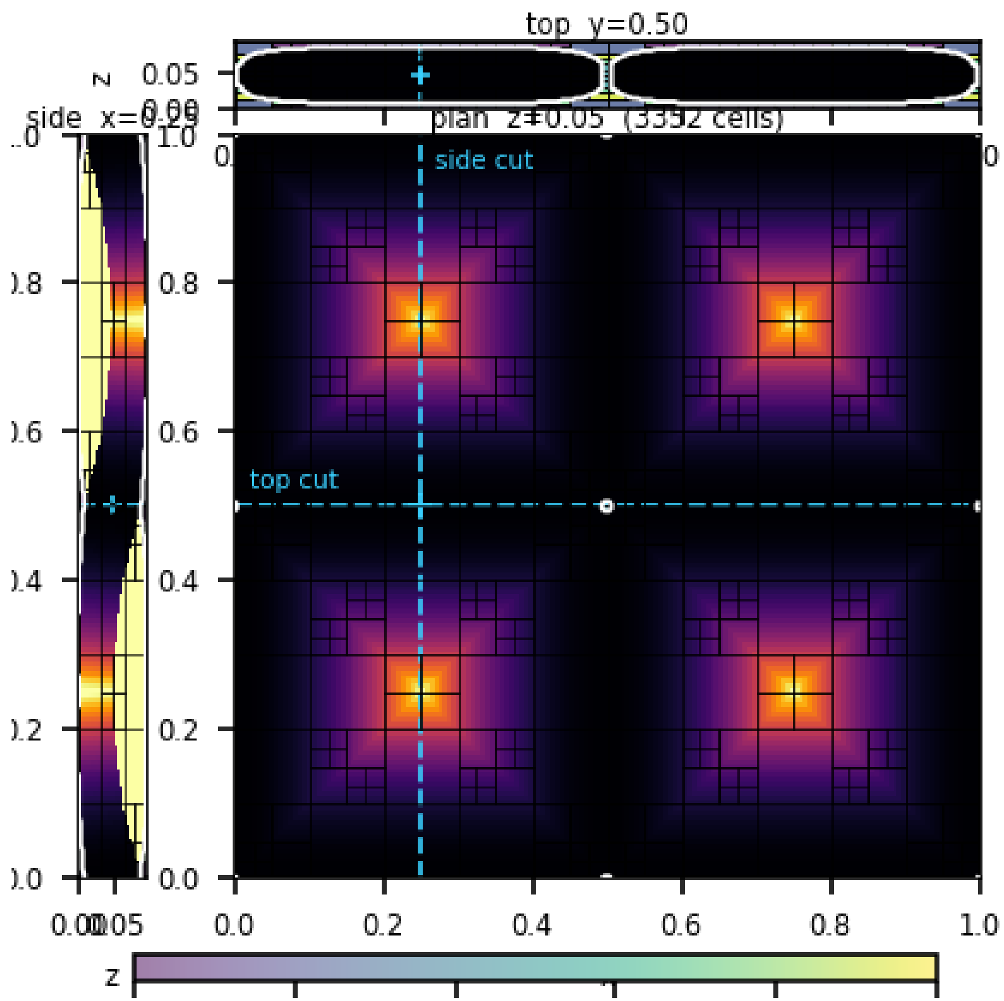
Micromechanics

matrix (matrix)	E = 3.00 GPa, nu = 0.35
fibre (fibre)	E_L = 230.00, E_T = 15.00, G_LT = 15.00, nu_LT = 0.20
yarn model	chamis
yarn Vf range	nominal 0.55, max 0.90
Vf_b bundle/RVE (MC)	0.914
Vf_local in-tow	0.55–0.90 (μ=0.63)
Vf_avg RVE fibre	0.574

Analysis

backend	mfem-periodic
cell type	hexahedron
homogenization mesh	24 x 24 x 8
BC	periodic (3-axis MPC / saddle-point)
material sampling	local_cloud
sampling resolution	6
periodic tolerance	1e-08
homogenization AMR	on
AMR iterations	2
AMR threshold	0.2
AMR dof budget	200000
AMR figure (right)	on — base 10×10×3 hex, 2 pass(es), τ=0.2; cuts plan z=0.05, top y=0.50, side x=0.25





Engineering constants E = (58.5, 58.5, 8.5) GPa; G = (3.87, 3.30, 3.30); ν = (0.09, 0.36, 0.36)

C_{eff} **[GPa]**

11	60.29	6.54	3.50	−0.00	−0.00	−0.00
22	6.54	60.29	3.50	0.00	−0.00	−0.00
33	3.50	3.50	8.83	0.00	−0.00	−0.00
23	−0.00	0.00	0.00	3.30	−0.00	0.00
13	−0.00	−0.00	−0.00	−0.00	3.30	0.00
12	−0.00	−0.00	−0.00	0.00	0.00	3.87